

High-Density Digital RRAM-based Memory with Bit-line Compute Capability

Shady Agwa, Yihan Pan, Thomas Abbey, Alexander Serb, Themis Prodromakis

Centre for Electronics Frontiers, Electronics and Computer Science, University of Southampton, SO17 1BJ, UK

Introduction:

The AI revolution shows the ever increasing performance demands of AI applications like Deep Neural Networks DNNs which consist of tens of layers and do computations on tens of millions of data weights [1]. Conventional Von Neumann architectures are currently struggling to meet these emerging performance demands with deep memory hierarchies to bridge the processor-memory performance gap [2]. Emerging technologies (like RRAMs) have meanwhile shown a real promise to address the increasing challenges of the conventional computing technology. While the main direction of research is focussing on exploiting the analogue memory attributes of RRAMs specially for analogue computing crossbars [3], this paper focuses on a different perspective of building high-density and digital-friendly RRAM-based memory that is a good alternative to the SRAM-based Last-Level Caches LLCs. This digital RRAM-based memory with conventional 1T1R bit-cells is proposed to be an on-chip gigantic data reservoir, with much higher density than SRAMs, to bridge the memory gap. The paper also shows that the digital RRAM-based memory is capable of doing robust bit-line compute which opens the door for digital in-memory computing architectures that can mitigate the Von Neumann bottleneck while adopting RRAM's high-density promise. Unlike analogue RRAM crossbars, RRAMs' digital in-memory computing capability should inherit the large scalability and the fast time-to-market of the digital domain with less engineering effort for optimisation as there is no need any more to build DACs and ADCs.

Basic Approach and Description of Circuits:

The objective of this paper is to spotlight the digital-friendly promise of RRAM devices by leveraging its On/Off resistance ratio to represent only two logic states (0/1). The main idea is to use an RC discharging time constant to distinguish between the Low Resistance State LRS and the High Resistance State HRS. Assuming a reasonable but still modest 10x On/Off ratio, the conventional 1T1R memory bit-cells have a 10x difference of discharging rates for their bit-line rails. Fig. 1 illustrates an example of two different 1T1R bit-cells with 10x resistance ratio and equal parasitic capacitances. By the end of the LRS time constant, a differential voltage sense-amplifier can easily detect the two different states of the two bit-cells. While the BL of LRS bit-cell has discharged to logic 0, the BL of the HRS bit-cell still has enough charge to be detected as logic 1.

To prove the concept of the digital RRAM-based memory, an RRAM device model [4] was used to build 1T1R bit-cells with LRS = 6.6K Ohm and HRS = 168.4K Ohm at supply voltage 1V with about 25.5x On/Off ratio. A schematic implementation of 8x8 1T1R array was built using a 180nm commercial CMOS technology node as a proof of concept. Fig. 2 shows the 8x8 1T1R array with word-line drivers, pre-charging circuits, pre-discharging circuits and differential voltage sense-amplifiers. The 8x8 array was programmed to have a balanced ratio of LRS and HRS bit-cells in addition to parasitic capacitances for BLs and BLBs to mimic the effect of extra 120 word-lines. The schematic simulation in Fig. 3 shows 8 digital read operations of the 8 word-lines where the read data was correctly verified. Fig. 4 shows the different discharging rates of the LRS bit-cell versus the HRS bit-cell which the sense-amplifier easily detects to read logic 0 and logic 1. The read operation has two phases, during the first phase the BL is charged to VDD while BLb is discharged to GND. During the second phase, both BL and BLb are disconnected while the wordline enable signal WLE activates the access transistor of the 1T1R bit-cell. If the RRAM device is at LRS, the BL discharges quickly into BLb and the sense-amplifier detects logic 0. If it is at HRS, the BL does not discharge fast enough and the sense-amplifier detects logic 1.

By activating two word-lines simultaneously, the sense-amplifiers detect the AND operation between the two activated word-lines. If

one bit-cell of the two activated bit-cells (within the same column) is at LRS, the BL discharges quickly and the sense-amplifier detects logic 0. If (and only if) the two bit-cells within the same column have HRSs, the BL still does not discharge fast enough and the sense-amplifier detects logic 1. This bit-line AND-ing has lower HRS value in comparison to the normal read due to the resistance parallelism but it still keeps the On/Off ratio with one order of magnitude which is very sufficient to distinguish output logic 0 and 1 as illustrated by the simulation in Fig. 5. Unlike conventional 6T CMOS SRAM bit-cells, the 1T1R is a single-ended bit-cell. To achieve a full bit-line compute capability, two adjacent 1T1R bit-cells can be used to capture the data and its complementary so that the bit-line AND-ing generates the bitwise AND and NOR operations as shown in Fig. 6. Representing data by two adjacent 1T1R bit-cells still consumes less area than conventional 6T CMOS SRAM bit-cells. From AND and NOR outputs, all other logic and arithmetic operations can be generated at the bottom of the peripheral circuits [5].

Simulation Results:

Based on the schematic simulation of the 8x8 1T1R array, the read operation of a single bit-cell is about 38.8 fJ at 1V and 550 MHz. Consequently, it should cost about 1.24 pJ to read a digital 32-bit word which is very promising for 180nm technology. This number is expected to be slightly increased after adding more peripherals like address decoder and muxing circuits. However, there is still room for circuit-level optimisation regarding the peripheral circuits and sense-amplifiers. On the other hand, the bit-line compute operation (AND or NOR) consumes about 51.4 fJ/bit at the same previous operating conditions which means a 32-bit in-memory AND or NOR operation costs about 1.65 pJ which is very comparable to the read operation. The layout of the design's different components is in progress with more circuit-level optimisation to build a complete digital RRAM-based array that can be taped out.

Conclusion:

This paper presents a promising perspective for leveraging the compatibility and high-density of emerging technologies like RRAMs to build digital-friendly RRAM-based memory that is capable of doing bit-line compute. The simulations show a promising momentum not only for using RRAM devices to build digital on-chip alternatives to SRAM-based LLCs but also for supporting RRAM-based digital in-memory computing. By increasing on-chip storage capacity, cheap computing paradigms can be adopted like stochastic computing which can exploit the great benefits of low cost RRAM-based digital in-memory computing. The relatively high-cost of the RRAM write operations is a major challenge against the wide adoption of digital RRAM-based memories. However, RRAM-based LLCs can be used in "write-once & read-many" scenarios and the technology's viability will improve as cheaper, digital-friendly RRAM devices are developed.

References:

- [1] Norman P. Jouppi et al., "In-Datcenter Performance Analysis of a Tensor Processing Unit," in 44th Annual International Symposium on Computer Architecture (ISCA '17). Association for Computing Machinery, NY, USA, 1–12. DOI: <https://doi.org/10.1145/3079856.3080246>.
- [2] D. Patterson et al., "A case for intelligent RAM," in IEEE Micro, vol. 17, no. 2, pp. 34–44, March–April 1997, doi: 10.1109/40.592312.
- [3] F. Cai et al., "A fully integrated reprogrammable memristor-CMOS system for efficient multiply-accumulate operations," Nat Electron 2, 290–299 (2019). <https://doi.org/10.1038/s41928-019-0270-x>.
- [4] I. Messaris et al., "A Data-Driven Verilog-A ReRAM Model," in IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, vol. 37, no. 12, pp. 3151–3162, Dec. 2018, doi: 10.1109/TCAD.2018.2791468.
- [5] K. Al-Hawaj et al., "Towards a Reconfigurable Bit-Serial/Bit-Parallel Vector Accelerator using In-Situ Processing-In-SRAM," 2020 IEEE International Symposium on Circuits and Systems (ISCAS), 2020, pp. 1–5, doi: 10.1109/ISCAS45731.2020.9181068.

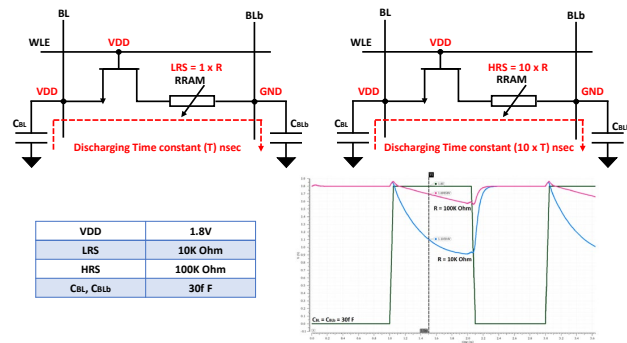


Fig. 1. An illustrating example of the main concept to read LRS and HRS bit-cells. Two conventional resistors are used to represent the two states and capacitors are added to BL and BLb to mimic the parasitics of tens of word-lines. BL is pre-charged to VDD and BLb is pre-discharged to GND. After only 0.5 nsec, the LRS discharges from 1.8V to 1.1V while the HRS only discharges to 1.7V. This voltage difference can be easily detected by a sense-amplifier.

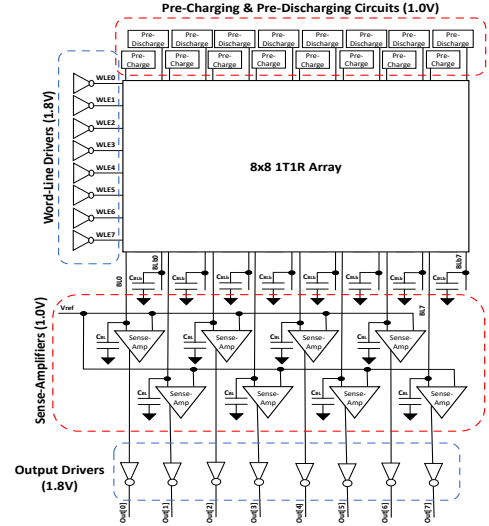


Fig. 2. 8x8 1T1R array setup using an RRAM model.

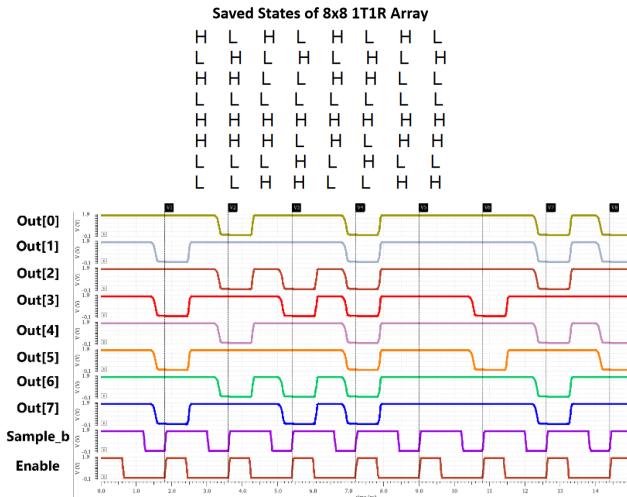


Fig. 3. Digital read simulation of 8x8 1T1R array with LRS bit-cells (Logic 0) and HRS bit-cells (Logic 1).

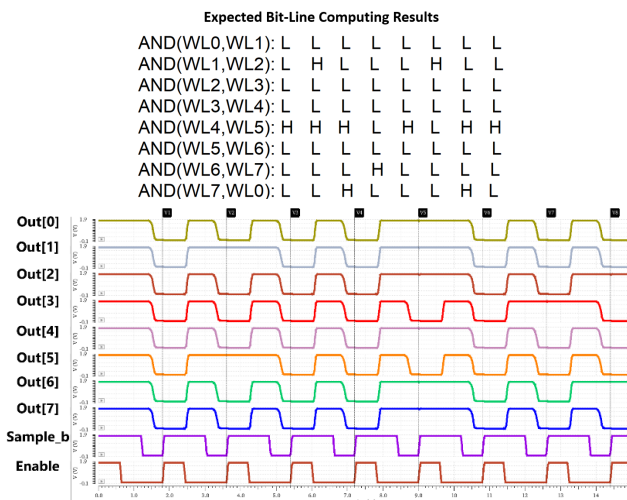


Fig. 5. Simulation of 8 digital bit-line computing operations (bitwise AND) for the 8x8 1T1R array.

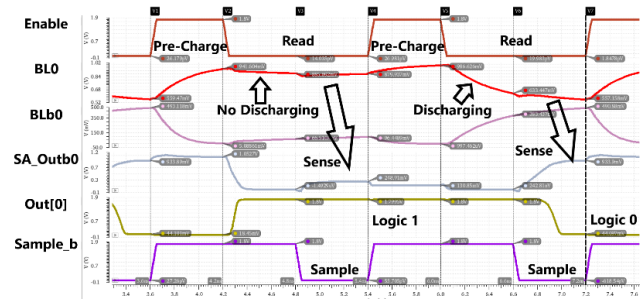
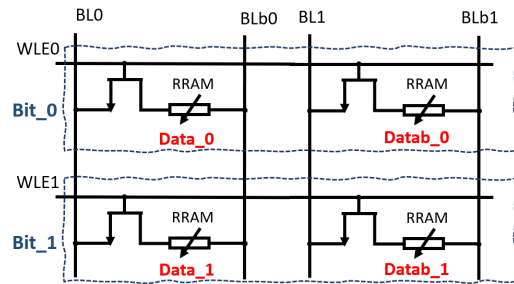


Fig. 4. Detailed simulation of HRS bit-cell (Logic 1) and LRS bit-cell (Logic 0). The BL is pre-charged to VDD and the BLb is pre-discharged to GND and then both of them are disconnected. The word-line enable signal is activated, so a discharging path from BL to BLb is created through the access transistor and the RRAM device. After a certain amount of time the sense-amp starts to detect Logic 1 for HRS and Logic 0 for LRS.



Data_0	Data_1	Datab_0	Datab_1	BL0	BL1
LRS	LRS	HRS	HRS	0	1
LRS	HRS	HRS	LRS	0	0
HRS	LRS	LRS	HRS	0	0
HRS	HRS	LRS	LRS	1	0

AND NOR

Fig. 6. AND & NOR operations using two 1T1R bit-cells.